SAFETY DATA SHEET
according to Regulation (EC) No. 1907/2006

SILK

ROYAUME DU PARFUM
INTERNATIONAL - PARIS

Version	Revision Date:	SDS Number:	Date of last issue: 28.01.2021
2.0	03.02.2021	300000880599	Date of first issue: 15.01.2021

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name	:	SILK
IFF Code	:	6023625

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture	:	Fragrance for consumer product

1.3 Details of the supplier of the safety data sheet

Company	:	IFF (NEDERLAND) BV ZEVENHEUVELENWEG 60 5048 AN TILBURG
Telephone	:	+31134642211
Telefax	:	+31134636032
E-mail address of person responsible for the SDS	:	sds@iff.com

1.4 Emergency telephone number

ORFILA (INRS): + 33 (0)1 45 42 59 59

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Skin irritation, Category 2	H315: Causes skin irritation.
Eye irritation, Category 2	H319: Causes serious eye irritation.
Skin sensitisation, Category 1	H317: May cause an allergic skin reaction.
Short-term (acute) aquatic hazard, Category 1	H400: Very toxic to aquatic life.
Long-term (chronic) aquatic hazard, Category 1	H410: Very toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms	:	
Signal word	:	Warning
Hazard statements	:	H315 Causes skin irritation. H317 May cause an allergic skin reaction. H319 Causes serious eye irritation. H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements :

Prevention:

P261 Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray.
P273 Avoid release to the environment.
P280 Wear protective gloves/ eye protection/ face protection.

Response:

P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P337 + P313 If eye irritation persists: Get medical advice/ attention.
P362 + P364 Take off contaminated clothing and wash it before reuse.

Hazardous components which must be listed on the label:

91-64-5	COUMARIN
138-86-3, 5989-27-5, 5989-54-8	limonene
115-95-7	linalyl acetate
33704-61-9	Dihydro pentamethylindanone
67874-81-1	OCTAHYDRO-METHOXY-TETRAMETHYL-METHANOAZULENE
144020-22-4	Reaction products of acetic anhydride and 1,5,10-trimethyl-1,5,9-cyclodecatriene
78-70-6, 126-91-0	Linalool
68155-66-8, 54464-57-2, 68155-67-9, 54464-59-4	Tetramethyl Acetyloctahydronaphthalenes
105-87-3	GERANYL ACETATE
1205-17-0	METHYLENEDIOXYPHENYL METHYLPROPANAL
4707-47-5	METHYL DIHYDROXY-DIMETHYLBENZOATE
1315251-11-6	5H-Cyclopenta[H]quinazoline, 6,7,8,9-tetrahydro-7,7,8,9,9-pentamethyl-
476332-65-7	Heptamethyl Decahydroindenofuran
127-51-5	Alpha-Isomethyl Ionone
104-54-1	CINNAMYL ALCOHOL
127-91-3	BETA-PINENES
106-22-9, 7540-51-4	Citronellol
82356-51-2, 63314-79-4	Methylcyclopentadecenone
80-56-8, 7785-26-4	pin-2(3)-ene
120-57-0	HELIOTROPINE
141-12-8	NERYL ACETATE
23696-85-7	ROSE KETONE-4
97-54-1	ISOEUGENOL

2.3 Other hazards

None reasonably foreseeable.

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This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran	1222-05-5 214-946-9 603-212-00-7 01-2119488227-29	Aquatic Chronic 1; H410 Aquatic Acute 1; H400	>= 10 - < 20
vanillin	121-33-5 204-465-2 01-2119516040-60	Eye Irrit. 2; H319	>= 10 - < 20
coumarin	91-64-5 202-086-7 01-2119943756-26	Acute Tox. 4; H302 Skin Sens. 1B; H317	>= 1 - < 10
limonene	138-86-3 205-341-0 601-029-00-7	Flam. Liq. 3; H226 Skin Sens. 1B; H317 Asp. Tox. 1; H304 Aquatic Chronic 1; H410 Skin Irrit. 2; H315 Aquatic Acute 1; H400 M-Factor (Acute aquatic toxicity): 1 M-Factor (Chronic aquatic toxicity): 1	>= 2,5 - < 10
linalyl acetate	115-95-7 204-116-4	Skin Irrit. 2; H315 Skin Sens. 1; H317 Eye Irrit. 2; H319	>= 1 - < 10
1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one	33704-61-9 251-649-3 01-2119977131-40	Aquatic Chronic 2; H411 Skin Sens. 1B; H317 Eye Irrit. 2; H319 Skin Irrit. 2; H315	>= 2,5 - < 10
3,7-dimethylnona-1,6-dien-3-ol	10339-55-6 233-732-6 01-2119969272-32	Eye Irrit. 2; H319 Skin Irrit. 2; H315	>= 1 - < 10
[3R-(3α,3aβ,6β,7β,8aα)]-octahydro-6-methoxy-3,6,8,8-	67874-81-1 267-510-5	Skin Sens. 1B; H317 Aquatic Acute 1;	>= 2,5 - < 10

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tetramethyl-1H-3a,7-methanoazulene	01-2120228335-61	H400 Aquatic Chronic 1; H410	
Reaction products of acetic anhydride and 1,5,10-trimethyl-1,5,9-cyclodecatriene	144020-22-4 482-330-9 01-0000020172-83	Skin Sens. 1B; H317 Aquatic Chronic 1; H410 Aquatic Acute 1; H400 M-Factor (Acute aquatic toxicity): 1	>= 2,5 - < 10
(+/-) trans-3,3-dimethyl-5-(2,2,3-trimethyl-cyclopent-3-en-1-yl)pent-4-en-2-ol	107898-54-4 411-580-3 603-150-00-0 01-0000015895-58	Skin Irrit. 2; H315 Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 1	>= 2,5 - < 10
linalool	78-70-6 201-134-4 01-2119474016-42	Skin Irrit. 2; H315 Skin Sens. 1B; H317 Eye Irrit. 2; H319	>= 1 - < 10
1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one	68155-66-8 915-730-3 01-2119489989-04	Skin Irrit. 2; H315 Skin Sens. 1B; H317 Aquatic Chronic 1; H410	>= 2,5 - < 10
geranyl acetate	105-87-3 203-341-5 01-2119973480-35	Skin Irrit. 2; H315 Skin Sens. 1; H317 Aquatic Chronic 3; H412	>= 1 - < 2,5
2-ethyl-3-hydroxy-4-pyrone	4940-11-8 225-582-5 01-2120758795-36	Acute Tox. 4; H302	>= 1 - < 10
α-methyl-1,3-benzodioxole-5-propionaldehyde	1205-17-0 214-881-6	Skin Sens. 1B; H317 Aquatic Chronic 2; H411 Repr. 2; H361 M-Factor (Acute aquatic toxicity): 1	>= 1 - < 2,5
2-phenylethanol	60-12-8 200-456-2 01-2119963921-31	Eye Irrit. 2; H319 Acute Tox. 4; H302	>= 1 - < 10
benzyl acetate	140-11-4 205-399-7 01-2119638272-42	Aquatic Chronic 3; H412	>= 1 - < 2,5
(Z)-3-hexenyl salicylate	65405-77-8 265-745-8 01-2119987320-37	Aquatic Acute 1; H400 M-Factor (Acute aquatic toxicity): 1	>= 1 - < 2,5

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methyl 2,4-dihydroxy-3,6-dimethylbenzoate	4707-47-5 225-193-0 01-2120762759-36	Skin Sens. 1B; H317	>= 1 - < 10
3-ethoxy-4-hydroxybenzaldehyde	121-32-4 204-464-7	Eye Irrit. 2; H319	>= 1 - < 10
5H-Cyclopenta[H]quinazoline, 6,7,8,9-tetrahydro-7,7,8,9,9-pentamethyl-	1315251-11-6 801-093-8 01-2120015081-77	Skin Sens. 1B; H317 Aquatic Chronic 1; H410 Aquatic Acute 1; H400 Acute Tox. 4; H302 STOT RE 2; H373	>= 1 - < 2,5
decahydro heptamethyl indenofuran	476332-65-7 449-360-4 01-0000018977-51	Skin Sens. 1B; H317 Aquatic Chronic 1; H410	>= 0,25 - < 1
3-methyl-4-(2,6,6-trimethyl-2-cyclohexen-1-yl)-3-buten-2-one	127-51-5 204-846-3	Aquatic Chronic 2; H411 Skin Sens. 1B; H317 M-Factor (Acute aquatic toxicity): 1	>= 0,25 - < 1
cinnamyl alcohol	104-54-1 203-212-3	Skin Sens. 1B; H317 M-Factor (Acute aquatic toxicity): 1	>= 0,1 - < 1
p-mentha-1,4-diene	99-85-4 202-794-6	Asp. Tox. 1; H304 Flam. Liq. 3; H226 Repr. 2; H361	>= 0,1 - < 1
(-)-pin-2(10)-ene	127-91-3 242-060-2 01-2119519230-54	Asp. Tox. 1; H304 Skin Irrit. 2; H315 Flam. Liq. 3; H226 Skin Sens. 1B; H317 Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 1	>= 0,25 - < 1
citronellol	106-22-9 203-375-0 01-2119453995-23	Skin Irrit. 2; H315 Skin Sens. 1B; H317 Eye Irrit. 2; H319	>= 0,1 - < 1
3-methyl-5-Cyclopentadecen-1-one	82356-51-2 429-900-5 01-0000017618-62	Skin Sens. 1B; H317 Aquatic Acute 1; H400 Aquatic Chronic 1; H410	>= 0,25 - < 1
isopentyl salicylate	2050-08-0	Aquatic Chronic 1;	>= 0,25 - < 1

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	218-080-2 01-2119969444-27	H410 Acute Tox. 4; H302 Aquatic Acute 1; H400	
pin-2(3)-ene	80-56-8 201-291-9 01-2119519223-49	Flam. Liq. 3; H226 Asp. Tox. 1; H304 Skin Irrit. 2; H315 Skin Sens. 1B; H317 Aquatic Acute 1; H400 Aquatic Chronic 1; H410 Acute Tox. 4; H302 M-Factor (Acute aquatic toxicity): 1	>= 0,1 - < 0,25
piperonal	120-57-0 204-409-7 01-2119983608-21	Skin Sens. 1B; H317	>= 0,1 - < 1
7-methyl-3-methyleneocta-1,6-diene	123-35-3 204-622-5 01-2119514321-56	Asp. Tox. 1; H304 Skin Irrit. 2; H315 Flam. Liq. 3; H226 Eye Irrit. 2; H319 Aquatic Chronic 2; H411 Aquatic Acute 1; H400	>= 0,1 - < 0,25
neryl acetate	141-12-8 205-459-2	Skin Sens. 1B; H317	>= 0,1 - < 1
1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one	23696-85-7 245-833-2	Skin Sens. 1A; H317 Aquatic Chronic 2; H411 Skin Irrit. 2; H315	>= 0,025 - < 0,1
isoeugenol	97-54-1 202-590-7	Acute Tox. 4; H312 Acute Tox. 4; H302 Skin Irrit. 2; H315 Eye Irrit. 2; H319 Skin Sens. 1A; H317 Acute Tox. 4; H332 STOT SE 3; H335	>= 0,01 - < 0,1

For explanation of abbreviations see section 16.

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SECTION 4: First aid measures

4.1 Description of first aid measures

General advice	:	Take Hazard and Precautionary phrases (section 2) into account.
If inhaled	:	Remove from exposure site to fresh air and keep at rest. If victim is unconscious, remove foreign bodies from the mouth. If victim has stopped breathing, give artificial respiration. Obtain medical advice.
In case of skin contact	:	Remove contaminated clothes. Wash thoroughly with water (and soap). Contact physician if symptoms persist.
In case of eye contact	:	Flush immediately with water for at least 15 minutes. Contact physician if symptoms persist.
If swallowed	:	Rinse mouth with water and obtain medical advice.

4.2 Most important symptoms and effects, both acute and delayed

Risks	:	Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation.

4.3 Indication of any immediate medical attention and special treatment needed

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media	:	Carbondioxide, dry chemical, foam.
Unsuitable extinguishing media	:	Do not use a direct waterjet on burning material.

5.2 Special hazards arising from the substance or mixture

Specific hazards during firefighting	:	Water may be ineffective.

5.3 Advice for firefighters

Further information	:	Standard procedure for chemical fires.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions	:	Avoid inhalation and contact with skin and eyes. A self-contained breathing apparatus is recommended in case of a

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major spill.
Prevent spreading over a wide area (e.g. by containment or oil barriers).

6.2 Environmental precautions

Environmental precautions	:	Keep away from drains, surface- and groundwater and soil.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up	:	Clean up spillage promptly. Remove ignition sources. Provide adequate ventilation. Avoid excessive inhalation of vapours. Gross spillages should be contained by use of sand or inert powder and disposed of according to the local regulations.

6.4 Reference to other sections

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling	:	Avoid excessive inhalation of concentrated vapors. Follow good manufacturing practices for housekeeping and personal hygiene. Wash any exposed skin immediately after any chemical contact, before breaks and meals, and at the end of each work period. Contaminated clothing and shoes should be thoroughly cleaned before re-use. If appropriate, procedures used during the handling of this material should also be used when cleaning equipment or removing residual chemicals from tanks or other containers, especially when steam or hot water is used, as this may increase vapor concentrations in the workplace air. Where chemicals are openly handled, access should be restricted to properly trained employees. Keep all heated processes at the lowest necessary temperature in order to minimize emissions of volatile chemicals into the air.
Advice on protection against fire and explosion	:	Keep away from ignition sources and naked flame.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers	:	Store in a cool, dry, ventilated area away from heat sources. Keep containers upright and tightly closed when not in use.

7.3 Specific end use(s)

Specific use(s)	:	No information available.

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SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Contains no substances with occupational exposure limit values.

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006:

Substance name	Environmental Compartment	Value
1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran	Fresh water	0,0044 mg/l
	Marine water	0,00044 mg/l
	Fresh water sediment	2 mg/kg dry weight (d.w.)
	Marine sediment	0,394 mg/kg dry weight (d.w.)
	Soil	0,31 mg/kg dry weight (d.w.)

8.2 Exposure controls

Engineering measures

Where appropriate, use closed systems to transfer and process this material.
If appropriate, isolate mixing rooms and other areas where this material is used or openly handled. Maintain these areas under negative air pressure relative to the rest of the plant.

Personal protective equipment

Eye protection : Use tight-fitting goggles, face shield or safety glasses with side shields if eye contact might occur.
Equipment should conform to EN 166

Hand protection
- Material : Nitrile rubber
- Break through time : > 60 min
- Glove thickness : 0,38 mm

- Material : Nitrile rubber
- Break through time : > 10 min
- Glove thickness : 0,1 mm

Avoid skin contact. Use chemically resistant gloves. The selected protective gloves have to satisfy the specifications of Regulation (EU) 2016/425 and the standard EN 374 derived from it. Be aware that in daily use the durability of a chemical resistant protective glove can be notably shorter than the break through time measured according to EN 374, due to the numerous outside influences (e.g. temperature). Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the

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danger of cuts, abrasion, and the contact time. The choice of an appropriate glove does not only depend on its material but also on other quality features and is different from one producer to the other.

Respiratory protection : Use local exhaust ventilation around open tanks and other open sources of potential exposures in order to avoid excessive inhalation, including places where this material is openly weighed or measured. In addition, use general dilution ventilation of the work area to eliminate or reduce possible worker exposures.
No respiratory protection is required during normal operations in a workplace where engineering controls such as adequate ventilation, etc. are sufficient.

If engineering controls and safe work practices are not sufficient, an approved, properly fitted respirator with organic vapor cartridges or canisters and particulate filters should be used:

a)while engineering controls and appropriate safe work practices and/or procedures are being implemented; or
b)during short term maintenance procedures when engineering controls are not in normal operation or are not sufficient; or
c)if normal operational workplace vapor concentration in the air is increased due to heat ;
d)during emergencies; or
e)if engineering controls and operational practices are not sufficient to reduce airborne concentrations below an established occupational exposure limit.

Protective measures : To the extent deemed appropriate, implement pre-placement and regularly scheduled ascertainment of symptoms and spirometry testing of lung function for workers who are regularly exposed to this material.
To the extent deemed appropriate, use an experienced air sampling expert to identify and measure volatile chemicals that could be present in the workplace air to determine potential exposures and to ensure the continuing effectiveness of engineering controls and operational practices to minimize exposure.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance : liquid

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Odour	:	conforms to standard
pH	:	not determined
Melting point	:	not determined
Boiling point	:	not determined
Flash point	:	75,00 °C Method: closed cup
Flammability (solid, gas)	:	not determined
Upper explosion limit / Upper flammability limit	:	not determined
Lower explosion limit / Lower flammability limit	:	not determined
Vapour pressure	:	0,14 hPa Calculated
Relative vapour density	:	not determined
Relative density	:	not determined
Density	:	not determined
Water solubility	:	not determined
Solubility in other solvents	:	not determined
Partition coefficient: n-octanol/water	:	not determined
Decomposition temperature	:	not determined
Viscosity, dynamic	:	not determined
Viscosity, kinematic	:	not determined
Oxidizing properties	:	not determined

9.2 Other information

No data available

SECTION 10: Stability and reactivity

10.1 Reactivity

No hazards to be specially mentioned.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : Presents no significant reactivity hazard, by itself or in contact with water. Avoid contact with strong acids, alkali or oxidizing agents.

10.4 Conditions to avoid

Conditions to avoid : Direct sources of heat.

10.5 Incompatible materials

Materials to avoid : Avoid contact with strong acids, alkali or oxidizing agents.

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10.6 Hazardous decomposition products

Carbon monoxide and unidentified organic compounds may be formed during combustion.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Not classified based on available information.

Product:

Acute oral toxicity : Acute toxicity estimate: > 2.000 mg/kg
Method: Calculation method

Skin corrosion/irritation

Causes skin irritation.

Components:

limonene:

Species : Rabbit
Exposure time : 24 h
Result : Skin irritation
Test substance : (undiluted)

Species : human
Exposure time : 48 h
Method : closed patch test
Result : No skin irritation

1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:

Species : human
Exposure time : 0,25 h
Method : EPISKIN Human Skin Model Test
Result : Skin irritation

Serious eye damage/eye irritation

Causes serious eye irritation.

Components:

vanillin:

Remarks : No data available

1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:

Species : Chicken eye
Method : OECD 438
Result : Eye irritation

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GLP	:	yes

Respiratory or skin sensitisation

Skin sensitisation

May cause an allergic skin reaction.

Respiratory sensitisation

Not classified based on available information.

<u>Components:</u>

limonene:

Test Type	:	LLNA
Species	:	Mouse
Result	:	Causes sensitisation.
Test substance	:	22% in ethanol/DEP (75:25)
Test Type	:	maximisation study
Species	:	human
Result	:	causes no sensitization
Test substance	:	20% in petrolatum

1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:

Test Type	:	LLNA
Species	:	Mouse
Method	:	OECD 429
Result	:	Skin sensitization
Test Type	:	repeated insult patch test
Species	:	Humans
Result	:	Does not cause skin sensitisation.
Test substance	:	10.% in ethanol/DEP (75:25)
Result	:	Does not cause respiratory sensitisation.

Germ cell mutagenicity

Not classified based on available information.

<u>Components:</u>

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Genotoxicity in vitro	:	Test Type: Ames test Metabolic activation: with and without metabolic activation Method: Mutagenicity (Escherichia coli - reverse mutation assay) Result: negative Test Type: Chromosome aberration test in vitro Metabolic activation: with and without metabolic activation Method: OECD 473 Result: negative

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coumarin:

Genotoxicity in vitro	:	Test Type: In vitro mammalian cell gene mutation test Test system: Chinese hamster ovary cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 476 Result: negative Test Type: reverse mutation assay Test system: Salmonella typhimurium Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative Test Type: Chromosome aberration test in vitro Test system: Chinese hamster ovary cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 473 Result: negative

Carcinogenicity

Not classified based on available information.

Reproductive toxicity

Not classified based on available information.

STOT - single exposure

Not classified based on available information.

STOT - repeated exposure

Not classified based on available information.

Repeated dose toxicity

Not classified based on available information.

Components:

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Species	:	Rat, male and female
NOAEL	:	>= 150 mg/kg
Application Route	:	Oral
Exposure time	:	90-day
Number of exposures	:	1x /day
Method	:	OECD 408

Aspiration toxicity

Not classified based on available information.

SECTION 12: Ecological information

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12.1 Toxicity

Components:

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Toxicity to fish	:	LC50 (Lepomis macrochirus (Bluegill sunfish)): 0,452 mg/l Exposure time: 21 d Test Type: flow-through test Method: OECD Test Guideline 204 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 0,9 mg/l Exposure time: 48 h Test Type: semi-static test Method: OECD Test Guideline 202 GLP: yes
Toxicity to algae/aquatic plants	:	ErC50 (Pseudokirchneriella subcapitata (green algae)): > 0,854 mg/l Exposure time: 72 h Test Type: static test Method: OECD Test Guideline 201 EbC50 (Pseudokirchneriella subcapitata (green algae)): 0,723 mg/l Exposure time: 72 h Test Type: static test Method: OECD Test Guideline 201
Toxicity to fish (Chronic toxicity)	:	NOEC: 0,068 mg/l Exposure time: 36 d Species: Pimephales promelas (fathead minnow) Test Type: flow-through test Method: OECD 210 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: 0,111 mg/l Exposure time: 21 d Species: Daphnia magna (Water flea) Test Type: semi-static test Method: OECD 211

12.2 Persistence and degradability

Components:

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Biodegradability	:	Result: Not readily biodegradable. Biodegradation: 2 % Exposure time: 28 d Method: Modified Sturm Test Test Type: aerobic

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Inoculum: activated sludge
Concentration: 10 mg/l
Result: Primary biodegradation
Biodegradation: 33,8 %
Exposure time: 28 d
Method: OECD 301 B
Remarks: see user defined free text

Test Type: aerobic
Inoculum: activated sludge, adapted
Concentration: 10,97 mg/l
Result: Primary biodegradation
Biodegradation: 28,3 %
Exposure time: 28 d
Method: OECD 301 B
Remarks: see user defined free text

coumarin:

Biodegradability : Test Type: aerobic
Inoculum: activated sludge
Result: Readily biodegradable.
Biodegradation: 90 %
Exposure time: 28 d
Method: OECD Test Guideline 301F
GLP: yes
Remarks: REACH data

12.3 Bioaccumulative potential

Components:

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Exposure time: 28 d
Bioconcentration factor (BCF): 1.584
Method: OECD 305

Partition coefficient: n-octanol/water : log Pow: 5,3 (25 °C)
GLP: no
Remarks: REACH data

log Pow: 5,9 (25 °C)
Method: OECD Test Guideline 117
GLP: yes
Remarks: REACH data

coumarin:

Bioaccumulation : Remarks: No bioaccumulation is to be expected (log Pow <= 4).

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Partition coefficient: n-octanol/water : log Pow: 1,39 (25 °C)

12.4 Mobility in soil

Product:

Mobility : Remarks: No data available

Components:

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Distribution among environmental compartments : Koc: 24547, log Koc: 4,39

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher..

12.6 Other adverse effects

Product:

Additional ecological information : There is no data available for this product.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : Dispose of according to local regulations. Avoid disposing into drainage systems and into the environment.

Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: Transport information

14.1 UN number

ADR : UN 3082

IMDG : UN 3082

IATA : UN 3082

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14.2 UN proper shipping name

ADR	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (HEXAMETHYLINDANOPYRAN)
IMDG	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (HEXAMETHYLINDANOPYRAN)
IATA	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (HEXAMETHYLINDANOPYRAN)

14.3 Transport hazard class(es)

ADR	:	9
IMDG	:	9
IATA	:	9

14.4 Packing group

ADR

Packing group	:	III
Classification Code	:	M6
Hazard Identification Number	:	90
Labels	:	9
Tunnel restriction code	:	(-)

IMDG

Packing group	:	III
Labels	:	9
EmS Code	:	F-A, S-F

IATA (Cargo)

Packing instruction (cargo aircraft)	:	964
Packing instruction (LQ)	:	Y964
Packing group	:	III
Labels	:	Class 9 - Miscellaneous dangerous substances and articles

IATA_P (Passenger)

Packing instruction (passenger aircraft)	:	964
Packing instruction (LQ)	:	Y964
Packing group	:	III
Labels	:	Class 9 - Miscellaneous dangerous substances and articles

14.5 Environmental hazards

ADR

Environmentally hazardous	:	yes

IMDG

Marine pollutant	:	yes (HEXAMETHYLINDANOPYRAN)

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IATA (Passenger)
Environmentally hazardous : yes

IATA (Cargo)
Environmentally hazardous : yes

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Other regulations:

Take note of Directive 92/85/EEC regarding maternity protection or stricter national regulations, where applicable.

Take note of Directive 94/33/EC on the protection of young people at work or stricter national regulations, where applicable.

15.2 Chemical safety assessment

A Chemical Safety Assessment is not required for this substance.

SECTION 16: Other information

Full text of H-Statements

H226	Flammable liquid and vapour.
H302	Harmful if swallowed.
H304	May be fatal if swallowed and enters airways.
H312	Harmful in contact with skin.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H332	Harmful if inhaled.
H335	May cause respiratory irritation.
H361	Suspected of damaging fertility or the unborn child.
H373	May cause damage to organs through prolonged or repeated exposure.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.

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H412 : Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road; AICS - Australian Inventory of Chemical Substances; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

In December 2003, the National Institute for Occupational Safety and Health ("NIOSH") published an Alert on preventing lung disease in workers who use or make flavorings [NIOSH Publication Number 2004-110].
In August 2004, the United States Flavor and Extract Manufacturers Association (FEMA) issued a report entitled "Respiratory Safety in the Flavor Manufacturing Workplace".
Both of these reports provide recommendations for reducing employee exposure and for medical surveillance in the workplace. The recommendations in these reports are generally applicable to the use of any chemical in the workplace and you are strongly urged to review both of these reports.
The report published by FEMA also contains a list of "high priority" chemicals. If any of these chemicals are present in this product at a concentration >= 1.0% due to an intentional addition by IFF, the chemical(s) will be identified in this safety data sheet.

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According to Regulation (EC) No. 1907/2006 the information in this safety data sheet is based on the properties of the material known to IFF at the time the data sheet was issued. The safety data sheet is intended to provide information for a health and safety assessment of the material and the circumstances, under which it is packaged, stored or applied in the workplace. For such a safety assessment International Flavors & Fragrances holds no responsibility. This document is not intended for quality assurance purposes.

Classification of the mixture:		**Classification procedure:**
Skin Irrit. 2	H315	Calculation method
Eye Irrit. 2	H319	Calculation method
Skin Sens. 1	H317	Calculation method
Aquatic Acute 1	H400	Calculation method
Aquatic Chronic 1	H410	Calculation method

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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